

Social Media Engagement Analysis Using Python

1. Introduction

Social media platforms generate vast amounts of user engagement data such as likes, comments, shares, impressions, and watch time. Analyzing this data helps in understanding user behavior, identifying trends, and improving content strategies.

This project focuses on analyzing a dataset of 5000 social media posts to extract meaningful insights using Python libraries such as Pandas, NumPy, Matplotlib, Seaborn, and Plotly.

2. Objective

- To clean and preprocess social media engagement data
 - To perform exploratory data analysis (EDA)
 - To visualize engagement patterns
 - To generate actionable insights on content performance and user behavior
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3. Dataset Overview

The dataset contains **5000 records** and **19 columns**, including:

- User details: age, gender, country
 - Post details: post type, category, hashtags, sentiment
 - Engagement metrics: likes, comments, shares, impressions, watch time, engagement rate
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4. Data Preprocessing

4.1 Handling Missing Values

- Missing values were identified using `isnull()`
- Numerical columns were filled using grouping up and median
- Categorical columns were filled using mode

4.2 Duplicate Handling

- Duplicate records were checked and no duplicates found

4.3 Data Formatting

- Converted date column to datetime format
- Standardized categorical values (gender, sentiment)
- Corrected unrealistic values in engagement metrics

4.4 Feature Engineering

- Extracted hashtag count
 - Created new feature: **engagement_score**
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5. Exploratory Data Analysis

5.1 Central Tendency

- Mean, median, and mode were calculated
- Likes, comments, and shares showed similar distributions
- Engagement rate showed high skewness

5.2 Dispersion

- Standard deviation and variance were calculated
- Follower count showed the highest variability

5.3 Skewness

- Most variables were nearly symmetric

- Engagement rate showed extreme positive skew

5.4 Kurtosis

- Most features were platykurtic (flat distribution)
 - Engagement rate showed very high kurtosis, indicating outliers
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6. Data Visualization

Matplotlib Visualizations

- **Scatter plot: Likes vs Impressions**

No strong correlation between likes and impressions High impressions don't always lead to high likes Data is widely scattered → high variability Possible clustering around specific like values Engagement is inconsistent across posts

- **Line chart: Daily engagement trend**

Engagement fluctuates over time No consistent increasing or decreasing trend Peaks indicate high-performing content days Drops show low engagement periods Overall engagement is dynamic

- **Bar chart: Posts by category**

The distribution is quite balanced (all categories are between ~590–680), so there is no extreme drop, but fitness and tech clearly lead, indicating audience preference or posting priority in those areas.

- **Pie chart: Gender distribution**

Male users form the largest group (37%) Female and other categories are nearly equal (~31%) Distribution is fairly balanced Indicates diverse user base No strong dominance by a single gender

- **Histogram: Age distribution**

Your audience is evenly distributed across age groups, with a small peak in the mid-age segment (30-40), which could be the most engaged demographic.

Seaborn Visualizations

- **Count plot: Post type distribution**

All post types are equally distributed. However, reels are even higher.

- **Bar plot: Average likes by category**

The likes count is evenly distributed showing that all the categories are equally liked by the followers

- **Violin plot: Followers vs sentiment**

There is no strong correlation between sentiment and follower count — accounts with positive, negative, or neutral sentiment have comparable audience sizes.

- **Pair plot: Numeric feature relationships**

Engagement metrics (likes, comments, shares) move together strongly Follower count is not a reliable predictor of engagement Watch time behaves independently Engagement is driven by a few high-performing posts

- **Heatmap: Correlation matrix**

Remove redundant variables (likes/comments/shares) Engagement is not driven by follower count Watch time does not translate into engagement Most features have weak predictive power individually

Plotly Visualizations

- **Interactive scatter plot**

All post types (image, reel, text, video) are evenly distributed, with no format dominating. High impressions do not always translate into high likes, indicating weak conversion. This suggests content effectiveness (not reach alone) drives likes.

- **Interactive line chart**

Engagement appears consistently spread across all dates, with no obvious upward or downward pattern.

7. Key Insights

7.1 Content Performance

Engagement is almost identical across all post types (image, reel, text, video), around ~30K. The overlapping error bars suggest no statistically significant difference between formats. This indicates post type is not a strong driver of engagement in your dataset. Engagement is likely influenced more by content quality, audience, or timing rather than format. All categories show very similar engagement (~29K–32K), with no clear standout. Slightly higher values for food, travel, lifestyle, and music, but differences are minimal. Brazil and Australia show the highest engagement rates, while USA and India are on the lower end. There is moderate variation across countries, unlike previous charts where differences were minimal. Large error bars (especially for Brazil) indicate high variability within countries. Overall, location seems to have some influence on engagement, but not strongly consistent.

7.2 User Trends

Engagement levels are highly variable across all age groups, from low to very high. No specific age group consistently outperforms others. Non-verified users show slightly higher engagement than verified users, but the difference is very small.

7.3 Behavioral Insights

Mobile shows slightly higher watch time compared to tablet and desktop. However, the differences are very small and closely clustered (~4000 sec).

7.4 Sentiment Analysis

Positive and negative sentiments show slightly higher engagement than neutral posts. However, the differences are very small (~30K range) and closely clustered.

8. Conclusion

The analysis consistently points to one central truth: **engagement is not driven by any single obvious factor**. Neither post type, category, age group, sentiment, nor follower count shows a strong or reliable influence on performance. Even widely assumed drivers—like impressions, watch time, or audience size—fail to translate consistently into higher likes or engagement. The data is highly variable, with performance spikes coming from specific posts rather than predictable patterns. This suggests that **content quality, relevance, and timing** play a far more critical role than structural factors like format or demographics. In other words, success in this dataset appears to be **content-driven rather than system-driven**, where a few high-performing posts disproportionately shape overall engagement trends.

At the same time, the audience itself is **diverse and evenly distributed**, with balanced representation across gender, age groups, and content categories. While some regions like Brazil and Australia show slightly higher engagement, the variation is inconsistent and not strong enough to form a clear strategic focus. Similarly, behavioral factors such as device usage and sentiment show only marginal differences. Taken together, this indicates that **there is no single “ideal audience” or “perfect content formula”**. Instead, growth and engagement likely depend on continuous experimentation, identifying what resonates on a case-by-case basis, and capitalizing on high-performing content patterns when they emerge.

9. Future Scope

- Apply machine learning models to predict engagement

- Perform deeper sentiment analysis using NLP
- Build dashboards for real-time analytics

10. Tools & Technologies Used

- Python
- Pandas, NumPy
- Matplotlib, Seaborn, Plotly